MILITARY SPECIFICATION

GRINDING MACHINES, JIG, VERTICAL, CNC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers floor mounted, vertical spindle, computer numerically controlled, precision, jig grinding machines with equipment and accessories specified herein.

1.2 Classification. The machines shall be of the following types, styles, and sizes. The specific type, style, and size required for a procurement shall be as specified (see 6.2.1).

Style A - Manual Data Input (MDI)
Style B - Standard Computer Numerical Control (CNC) with MDI Capabilities

Type I - Single Column

** Size	Table Travel (inches)
1	16 X 10
2	18 X 11

Type 11 - Double Column

** Size	Table Travel (inches)
1	25 X 18
2	32 x 24
3	36 X 24
4	48 X 24
5	48 x 24 (Extended table size)

** When another size is required, machine size and corresponding characteristics shall be as specified (see 6.2.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3415

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and Standards. The following specifications and standards form a part of this specification to th extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified> the issue of documents not listed in the DODISS shall be the

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issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standard for.

ANSI B7.1 - Abrasive Wheels, Use, Care, and Protectionof, Safety Codes for the.

ANSI B11.9 - Safety Requirements for the Construction, Care and Use of Grinding Machines.

ANSI B74.13 - Marking for Identifying Grinding Wheels and Other Bonded Abrasive

ANSI B74.2 - Shapes and Sizes of Grinding Wheels, and Shapes, Sizes, and Identification of Mounted Wheels, Specification for.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION

AGMA 360.02 - Manual For Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association Standards Department, Sutie 1000, 1901 North Fort Myer Drive, Arlington, VA 22209-1695.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

ISO R1122 - Gears, Glossary of, Geometrical Definitions.

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(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA) STANDARDS

RS-227	– One Inch Perforated Tape.
RS-232	– Interface between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange.
RS-244	– Character Code for Numerical Machine Tool Control Perforated Tape.
RS-267	– Axis and Motion Nomenclature for Numerically Controlled Machine Tools.
RS-274	- Interchangeable Variable Block Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines.
RS-281	– Construction Standards - Numerical Machine Tool Control.
RS-358	– Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape.

(Application for copies should be addressed to the Electronic Industrial Association, Engineering Department, 2001 Eye Sheet, N.W,, Washington, DC 20006 l)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design. The machine shall be new and one of the manufacturer’s current models capable of locating and grinding cylindrical and conical shaped holes and combinations of radii, tangents, angles, and flats in hardened

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steel and other metals and materials in accordance with the requirements herein. Style A-machines shall be manual data input (MDI) machines with two axes simultaneous control; style B machines shall be computer numerically controlled machines (CNC) with MDI capabilities and three, four or five axes simultaneous control. Type I machines shall be of single column construction; type 11 machines shall be of double column construction. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.1.1 Measuring systems. Unless otherwise specified (see 6.2,1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.1.1.1 Measuring and indicating device calibrations. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments , and dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.1.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine’s intended use or performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.1.3 Energy efficiency. The machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.1.4 Operating controls. All operating controls necessary for manual or NC operation of the machine shall be mounted on either a control console or pendant stand clearly identified to facilitate its use. Voltage and current meters, indicator lights, and all other operational control aids shall be

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arranged and mounted convenient to and in full view of the operator. An emergency stop switch shall be furnished on the control station.

3.1.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with ANSI B11.9 and with all of the requirements of OSHA 2206 that are applicable to the machine itself. In the event of a conflict between the requirements of the ANSI and OSHA standards, the OSHA standards shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.1.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.1.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or In the machine.

3.1.8 Environmental protection. The machine shall be designed and constructed such that, under the operating, service, transportation and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state or local statutes in effect at the point of the installation.

3.1.9 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification to the part or the machine.

3.2 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be cable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.2.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on machine. Such processes may be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.2.2 Welding, brazing- or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

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3.2.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.2.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.5 Painting. Unless otherwise specified (see 6.2,1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.2.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section reference therein.

3.2.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position. .

3.2.8 Gears. All gears and pinions used in the machine or its components, shall be properly selected for the intended application and fabricated to either English (USA) standards per AGMA 390.03 and AGMA 360.02 or Metric (SI) standards per the International Organization for Standardization, ISO-R-1122 and 1SO-54.

3.2.9 Machine ways. The machine ways shall be adequately protected whenever possible to exclude grinding dust, dirt, coolant, and other hazards that may cause damage or result in inaccurate functioning of any members. The machine ways shall be hardened and ground and suitable for the high grade precision numerically controlled machine covered by this specification.

3.2.10 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with the requirements of MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.3 Components. The machine shall consist basically of the following components. Any additional components necessary for the operation of the machine shall be furnished in accordance with manufacturer's standard practice.

3.3.1 Base. The base shall be a casting of iron, iron alloy or a fabricated weldment, ribbed and braced to minimize distortion, vibration, and

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deflection. The base shall support the column(s), bridge, spindle housing assembly, worktable, saddle, and cross carriage, as applicable. The coolant; hydraulic and. electrical systems shall be separate external units. When applicable, an access door(s) or removable plate(s) shall be provided to service units housed within. Guideways shall be provided to accommodate either a sliding saddle or worktable, as applicable. Means shall be provided to level the machine.

3.3.2 Column-assembly: Each column shall be a one-piece casting of box-like construction, Internally ribbed and braced to minimize distortion and deflection under normal loads. The column on type I machines shall accommodate the main spindle housing assembly and its associated components. Guideways shall serve to support and guide the spindle housing assembly through its full range of vertical adjustment. Means shall be provided to protect column ways from dirt. Means shall be -provided for vertical adjustment of the spindle housing assembly with a device to securely lock the spindle housing assembly in any location within its vertical range. A graduated scale or some other visible means shall be provided to return the spindle housing assembly to the same location after assembly movement. The machine shall have either stops or limit switches located at the extremes of spindle housing assembly motion to prevent excess travel of the spindle assembly. Type II machines shall meet the same basic requirements as the type I machines except that they shall have a double column, bridge and sliding cross carriage configuration accommodating the main spindle housing assembly.

3.3.3 Spindle -housing -assembly. The spindle housing assembly shall include, but not be limited to, the main drive spindle and the grinding spindle or head with corresponding feed and speed mechanisms, vertical slide handwheel, spindle housing clamp, adjustable depth stop, heating elements, and means for indicating housing temperature. The assembly shall be made of a nickel iron alloy with a low coefficient of thermal expansion. Means shall be provided in the spindle housing assembly to simulate bearing running heat when the spindle is not running but the machine is operative to minimize the effects of expansion and contraction due to temperature changes common with variations in spindle operation. The assembly shall be mounted on the column or cross carriage guideways, as applicable, and shall traverse vertically (W axis). The assembly shall be balanced and counterbalanced. A non-influencing clamping mechanism shall be provided to lock the assembly to the column at any position within its vertical travel. The main spindle shall be of steel alloy and shall rotate in preloaded ball or roller bearings. The motor that drives the grinding spindles or heads shall be either an electric or pneumatic type. Provisions shall be made to adjust the spindle speed in either continuous adjustment or in steps of not greater than 10 percent. The grinding head shall be designed to accommodate grinding wheels mounted on 3/8 and, when specified (see 6.2.1), 1/4, and 1/8 inch shanks using adapters, if required. Means shall be provided to indicate vertical spindle housing assembly location.

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3.3.3.1 Taper grinding. A taper grinding attachment shall be provided to allow grinding of convergent and divergent tapers. A taper-indicating dial or plate shall be provided to set the angle of inclination throughout its entire range. The indication dial shall have a maximum graduation of 30 minutes,

3.3.3.2 Radial grinding wheel-off-set. Provision shall be made for horizontal adjustment or off-setting (U axis) of the grinding wheel from the main spindle's fixed axis of rotation. A rough adjustment shall be provided to allow the grinding wheel to be offset to the approximate required dimension while the machine is inoperative. A fine adjustment shall be provided to adjust the grinding wheel to the exact dimension while the machine is in operation. When specified (see 6.2.1), U-axis grinding wheel off-setting shall be numerically controlled and the U-axis location shall be displayed on a universal display . The fine adjusting off-set mechanism, when not numerically controlled, shall have a dial to indicate offset with a maximum graduation of 0.0001 inches.

3.3.3.3 Axial reciprocation. Power reciprocation of the grinding spindle shall be provided. Manual operation, as required by the operator for set-up, will be through the use of a handwheel conveniently located near the operator. When specified (see 6.2.1), Z-axis feed shall be numerically controlled and location indicated on a universal display. Power vertical reciprocation of the grinding spindle shall be provided with a pneumatic compensating device or counterbalance. An adjustable positive stop shall be provided to set lower travel limit to prevent overtravel. A dial or scale shall indicate spindle position, graduated in increments not larger than 0.001 inches, when not numerically controlled.

3.3.3.4 Contour-grinding. Provision shall be made to grind cylindrical sections between predetermined angular limits. It shall be possible to swing the spindle back and forth through a controlled arc. A direct reading indicator or scale shall be provided. Adjustable stops shall be provided so that an arc of any desired size within the range of the machine can be established without danger of overtravel. A mechanism to swing the spindle manually through the arc defined by the stops shall be provided.

3.3.3.5 Planetary -movement-of-grinding-wheel-(C-axis). Provision shall be made for planetary movement (C axis) of the grinding wheel. The grinding wheel shall not only rotate about its own axis but shall have circular or planetary movement so that the eccentrically offset wheel follows around the contour or hole to be ground. Planetary offsetting shall be numerically controlled with C-axis location indicated on a universal display.

3.3.4 Worktable. The worktable shall be a casting of iron or iron alloy, designed and proportioned to withstand any loads that may be encountered during operation at its maximum capacity. Unless otherwise specified (see 6.2.1), the number, size and configuration of T-slots in the table shall be manufacturer's standard. The surfaces of the worktable shall be hand or machine finished for a work mounting surface. When flood

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coolant is provided (see 3.3.10), the top of the table shall have coolant troughs or similar means for draining spent coolant into its reservoir. The table shall be provided with a means of securing workplaces in place to permit precision grinding without drifting. The worktable shall be mounted on the traversing saddle on type I machines and shall travel in the X-axis direction. On type II machines, the worktable shall be mounted on the base and shall traverse in the X-axis direction. The worktable shall be provided with the necessary protective shields or covers, or both, for leadscrews, ballscrews, motors, and guideways to guard against the entry of foreign material. Means shall be provided to prevent table movement after positioning. Stops shall be provided at each end of travel to prevent table overtravel. A digital display shall indicate worktable position (X and Y axes). Worktable functions on all machines shall be numerically controlled.

3.3.4.1 Precision-rotary-table-(A/D -axes). When specified (see 6.2.1), the machine shall be equipped with a 360,000 position precision rotary table (D axis) or precision tilting, rotary table (A/D axes). The A axis represents the vertical rotating axis of the rotary table and the D axis represents the horizontal rotating axis of the table. The table operation shall be numerically controlled. The rate of rotation of the rotary table shall be manufacturer's standard. The size of the rotary table shall be as specified (see 6.2.1). For tilting rotary tables (A axis), a tailstock shall be provided when specified (see 6.2.1). The rotary table shall meet the accuracy and alignment requirements specified in table II.

3.3.5 Saddle. Type I machines shall be provided with a saddle. The saddle shall be a casting of iron or iron alloy designed and proportioned to withstand maximum loads which may be imposed during operation of the machine. The saddle shall be mounted on the base guideways and shall traverse in the Y-axis direction. Longitudinal guideways shall be provided on the saddle top to accept a sliding worktable. Means shall be provided to prevent movement of the saddle after positioning. Stops shall be provided at each end to prevent overtravel. A digital display unit shall be provided to indicate saddle position. The saddle slide shall be secured in place over the entire saddle travel range by either servo devices or comparable means as directed by the control. Saddle functions shall be numerically controlled. Covers, protective shields, or both, shall be provided for leadscrews, motors, and guideways, as applicable, to guard against the entry of foreign matter.

3.3.6 Bridge. Type 11 machines shall be constructed with a cast iron, bridge-like member. It shall be internally ribbed and braced to minimize distortion and deflection experienced under maximum design loads. The bridge shall be fixed, supported on two vertical box-type columns located on either side of the base casting, simulating a bridge over the base and worktable. Horizontal guideways shall be provided on the bridge to accept a moveable cross carriage assembly and to allow the cross carriage to move in the Y-axis direction. On machines with a cross rail, vertical guideways shall be provided in the columns supporting the cross rail to allow the

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cross rail to traverse in the vertical (W-axis) direction. The cross rail shall be equipped with fully automatic, distortion-free clamping. Horizontal guideways shall be provided on the cross rail to accept a cross carriage and to allow the cross carriage to move in the Y-axis direction. The cross rail shall be pneumatically balanced to assure parallel movement of the spindle headstock in relation to the worktable. Covers or protective shields shall be provided to guard against the entry of foreign material to leadscrews, ballscrews, motors, and guideways. Stops shall be provided to prevent overtravel.

3.3.7 Cross-carriage. Type II, machines shall be provided with a cross carriage. The cross carriage shall be made of iron or iron alloy, precision fitted to the bridge or cross rail, traversing in the Y-axis direction. The cross carriage on machines without a cross rail shall be provided with vertical guideways to accommodate the spindle housing assembly and its associated components and to allow the assembly to traverse ‘vertically (W-axis). On machines with a cross rail, the cross carriage and spindle housing assembly shall travel in the Y-axis direction on horizontal guideways on the bridge and vertically (W-axis) on vertical guideways in the columns. Aprons or covers shall be provided to guard the guideways from contamination. Means shall be provided to secure cross carriage location after positioning. The cross carriage shall be provided with power drives.

3.3.8 Grinding-wheels. Unless otherwise specified (see 6.2.1), grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1.

3.3.9 Lubrication. A lubrication system shall be provided which will insure adequate lubrication for all moving parts. The system shall deliver lubrication to all leadscrews, ballscrews, mating gears, sliding parts, and bearings except the sealed-for-life type. Each lubricant reservoir shall have a means for determining and maintaining fluid level. The lubricant reservoir shall have sufficient capability to operate for a minimum of 80 hours of feed time without being serviced. The lubrication system shall have a device for indicating oil level and shall be readily visible to the operator. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. A recommended preventive maintenance schedule shall be provided.

3.3.10 Coolant system. Unless otherwise specified (see 6.2.1), the machine shall be supplied with a complete coolant system. For type 11 machines, the coolant system shall be numerically controlled with a selection of coolant type and on/off of coolant flow. The system shall have adequate capacity for all types of grinding operations for which the machine is designed and shall have means for regulating the flow of flood coolant or for adjusting coolant supply, as applicable. All components necessary to make the system completely functional shall be furnished including reservoirs, splash guards, filters, drain troughs, and clean-out plates.

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3.3.11 Hydraulic system. The hydraulic system, when supplied, shall be complete with pumps, valves, piping, cylinders, reservoir, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. The system shall include filters to insure delivery of clean fluid. The reservoir shall have a sight gage to indicate fluid level. The fluid temperature shall be maintained within the proper operating range during all normal operations of the machine.

3.3.12 Pneumatic system. Unless otherwise specified (see 6.2.1), a pneumatic system shall be provided. Shop air shall be provided having a line pressure of 90 to 120 pounds per square inch (psi). The machine shall have an air line filter, water trap, and oil mist lubricator, as necessary. The system shall include all devices such as pressure regulators, pressure reducers, solenoid valves, and check valves that are necessary to meet the requirements specified herein. Air pressure gauges shall be provided to indicate input air pressure in all lines that are downstream from a pressure control device.

3.3.13 Electrical system; Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), the machine shall be operable on a 440 volts (with an allowable line fluctuation of +10 percent), three phase, 60 hertz (Hz) source and shall be initially wired for operation on 440 volts. It shall be possible to operate all electric equipment on the machine after connection to a single power source.

3.3.13.1 Motors. Motors shall be rated for continuous duty and shall have ball or-r bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for dripproof enclosure. Electric motors shall be either alternating current (AC), 3 phase,60 Hz or industrial direct current (DC) motors conforming to ANSI/NEMA MG-1, equipment with solid state rectifier. Variable speed DC spindle motors shall develop rated horsepower through at least 90 percent of the range of spindle speeds available from a gear train or direct coupling. Pneumatic motors shall be of the vane or impulse reaction turbine air motor type controlled by the regulation of air pressure (see 3.3.12).

3.3.13.2 Control-circuit ‘voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 115 volts.

3.3. 13.3 Main electrical-disconnect. A manually operated fusible disconnect twitch or circuit breaker of adequate size to carry the entire electrical load of the equipment shall be provided for connection to the primary power source specified.

3.3.14 Numerical-control. The machine offered shall be completely programmable through a microprocessor type computerized numerical control (CNC)

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as described in 3.3.14 through 3.3.14.36. Style A machines shall be two axes programmable machines with linear interpolation and manual data input (MDI). Style B machines shall be either three, four or five axes programmable machines with full linear and circular interpolation. Parabolic interpolation shall be available when specified (see 3.3.14.2). Style B machines shall have either tape, cassette, diskette or manual data input capabilities. The control unit shall conform to EIA Standard RS-281. A line voltage of +10 percent from normal shall not adversely affect the numerical control function. The control shall function in ambient temperatures ranging from 50° to 120°F and relative humidities ranging from 5 percent to 95 percent non-condensing. The control shall automatically shut down when internal operating temperature exceeds safe operating temperature. Control resolution shall be 0.0001 inch for each linear axis and not more than 0.001 of a degree for each rotary axis. The axes shall be identified in accordance with EIA Standard RS-267. The control shall initiate an error or halt signal should a fault condition occur in the control.

3.3.14.1 Manual -data -input -(MDI) . Style A machines shall be two axes programmable machines with linear interpolation and MDI. The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, two axis feedrate control of X and Y axes, and other part program directed functions for jig grinders of the type described herein. The control shall have linear interpolation and simultaneous control of two axes. Control features shall include buffer storage, part program storage, part program edit capability, and control diagnostics. The MCU shall direct machine functions from part program data stored in memory or manually input through a keyboard.

3.3.14.2 Computer numerical control (CNC) with MDI capabilities. Style B machines shall be completely programmable through a microprocessor-type computerized numerical control (CNC). The MCU shall provide automatic control of machine functions, operating modes, and two axes (X and Y) positioning and feedrate control. A third axis shall provide positioning control of the planetary movement of the grinding wheel (C axis). An options?. fourth axis shall be available for a rotary table (A/D axis). The axes to be numerical ly controlled shall be as specified (see 6.2.1). When specified (see 6.2.1), a fifth axis shall be available for either radial infeed (U axis) or axial feed (Z axis) of the grinding wheel. The MCU shall also provide automatic control of rapid traverse rates, axis dwell. times, and shall initiate action should a machine fault occur. The control shall have linear and circular interpolation. Parabolic interpolation shall be provided when specified (see 6.2.1). The control shall be capable of three, four or five axes simultaneous control, as applicable. Control features shall include a programmable interface between the MCU and grinder, buffer storage, fixed cycles, part program storage, part program and buffer edit capability, and control diagnostics. The MCU shall direct machine functions from part program data stored in memory, through manual data input (MDI), or from data directly input from the input media specified in 3.3.14.3. All necessary executive diagnostic and interface program routines shall be furnished in the form of one inch, eight channel binary coded decimal punched tape or

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other input form as appropriate for input the media specified for controls having volatile memory.

3.3.14.3 Data input. Unless otherwise specified (see 6.2.1), the input media for the control on style B machines shall be one inch, eight channel punched tape. The data input format and codes shall comply with EIA Standards RS-274 and RS-358 (ASCII) or, when specified (see 6.2.1), RS-244 (EIA) in lieu of RS-358 (ASCII). The control shall be equipped with a photoelectric tape reader capable of holding, feeding, and reading tape prepared in the format specified at a rate of not less than 150 characters per second. Unless otherwise specified (see 6.2,1), the tape reader shall be supplied with standard feed and take-up spools 5-1/4-inch in diameter. The tape reader shall be housed in a dust free area of the numerical control cabinet and shall be accessible for cleaning and tape handling. The control shall continuously check for the following errors:

a. Improper parity.
b. Misaligned sprocket holes.
c. No sprocket holes.

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.3.14.4 Data-display. Style A machines shall be provided with a means to display the following: optical display of the mode of operation being used, digital readout of actual position and desired position for two axes (X and Y axes), block display, and control and system error messages. Style B machines shall be provided with an optical display capable of displaying at least 250 characters simultaneously. Unless otherwise specified (see 6.2.1), the display shall, as a minimum, show the following: block display, actual and desired position of each axis (as applicable), feedrate and override conditions, and control and system error messages. For both style A and style B machine data displays, position displays shall indicate a sign (positive or negative) and a minimum of 7 digits.

3.3.14.5 Control-system-modes of operation. The operational modes of the CNC control shall include manual, automatic, single block, MDI, sequence number search and optional stop.

3.3.14.6 Departure-control. The maximum departure permitted by a single command in either Incremental or absolute mode shall be not less than plus or minus 99.9999 inches. The control shall automatically control the acceleration and deceleration of the slides. The resolution of slide movement in either direction shall be not more than 0.0001 inch.

3.3.14.7 Absolute/incremental-input. The control shall have absolute/incremental Input capability. This mode shall be selected by a preparatory "G" code or setpoint switch which may be switched during the part program.

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3.3.14.8 Switchable inch/metric input. The control system shall provide switchable inch/metric input capability.

3.3.14.9 Part program storage. Unless otherwise specified (see 6.2.1), part programs shall be stored in either solid state or bubble type memories. Unless otherwise specified (see 6.2.1), the MCU on style A machines shall store in memory a minimum of the equivalent of 50 feet of postprocessed part program tape data and the MCU on style B machines shall store in memory the content of not less than 200 feet of postprocessed part program tape data.

3.3.14.10 Standby battery power. Unless otherwise specified (see 6.2.1), standby battery power shall be provided for controls having volatile memory to hold the fully loaded memory for a period of not less than 72 hours after the primary power source fails.

3.3.14.11 Part program-edit. A program edit capability shall be provided to allow block Insertion, deletion or modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.3.14.12 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from zero to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that programmed.

3.3.14.13 Sequence-number-search. The control shall permit the operator to search for any desired sequence number.

3.3.14.14 Mirror-image. Style B machines shall be capable of mirroring from the same input data through the inversion of X and Y axes. Mirroring in four quadrants shall be provided when specified (see 6.2.1). Inversion of each axis shall be by keyboard or switch entry. Mirror imaging shall not affect the directional sense of manual controls. Reverse settings shall be displayed on the data display.

3.3.14.15 Preparatory function (G-Codes). The preparatory function codes of the control shall be in accordance with EIA Standard RS-274.

3.3.14.16 Miscellaneous functions -(M-Codes). The miscellaneous function codes of the control shall be in accordance with EIA Standard RS-274.

3.3.14.17 Block delete. The control shall have a block delete feature to permit by-passing blocks of programmed data.

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3.3.14.18 Buffer storage. Style B machine controls shall have buffer storage for transferring command data from the tape reader to internal storage without delaying the next incoming command and without interrupting the machine functions. The internal storage shall store multiple blocks of data until required by the controlling devices.

3.3.14.19 Block-by-block-read. The control shall have a block-by-block read function to provide for reading information one block at a time.

3.3.14.20 Automatic-cycle. The control shall have the capability of running in automatic cycle. Once programmed, the control shall automatically run for an infinite number of parts.

3.3.14.21 Grinding-wheel-diameter -compensation. Compensation shall be provided for grinding wheel diameters differing from those utilized in programming. The grinding wheel diameter compensation shall have a range of not less than the following:

a. Inch system +0.0001" to +1.0000".
b. Metric system +0.001mm to +25.400mm.

3.3.14.22 Fixture compensation. A fixture compensation feature shall be provided to permit compensation for fixture position errors.

3.3.14.23 Reversal -error-compensation. Error resulting from reversing the direction of the leadscrew shall be minimized by the use of appropriate mechanical devices designed to eliminate as much reversal error as possible. Electronic compensation within the control unit may be used to reduce the effects of reversal error. Means shall be provided to adjust the anti-backlash device and to input new compensation values, if applicable, to correct for wear and usage during the life of the machine. The backlash eliminator/compensation system shall provide sufficient accuracy to meet the tolerances given in table II.

3.3.14.24 Axis -calibration. Error compensation shall be provided as necessary to correct repeatable errors determined to exist due to ballscrew and feedback inaccuracies. A means shall be provided to input new error compensation values as required.

3.3.14.25 Additional-features. Unless otherwise specified (see 6.2.1), the following additional numerical control features shall be provided:

a. Incremental and continuous jog for each axis.
b. Leading zero suppression or decimal point programming.
c. Inches per minute (IPM)/millimeters per minute (MMPM), programming in all modes of operation.
d. Data edit keylock.

3.3.14.26 Operator control panel. Unless otherwise specified (see 6.2.1), the control unit shall have a control panel which provides at least

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the following control functions: Emergency Stop; Cycle Start; Optional Stop; Mode Selection; Feed Hold; Feedrate Override; Spindle Speed Override; and EOB Stop.

3.3.14.26.1 Pendant control. When specified (see 6.2.1), a pendant or moveable stanchion control shall be furnished in lieu of an operator control panel. The pendant control shall be constructed to enable the operator to control machine functions while in close observation position. Unless otherwise specified (see 6.2.1), the pendant control shall be capable of the same functions as stated for the operator control panel in paragraph 3.3.14.26.

3.3.14.27 Reference-zero. The control unit shall have a reference zero to move-each axis to a reference point to establish synchronization and reset all registers to zero.

3.3.14.28 Set zero. The control unit shall have a “set zero” feature to permit the specified axis register to be reset to zero establishing present axis location as zero.

3.3.14.29 Program tryout. The control shall have a program tryout feature to permit new part programs to be tested through the control without machine motions.

3.3.14.30 Operational-software. Operational software shall be resident in the control and shall contain all logic necessary to effectively operate the machine tool. The software shall provide for meaningful diagnostics relating to the utilization of the control, machine tool, part program, and operator’s actions. On style B machines, all necessary executive diagnostic and interface program routines for those controls having volatile memory shall be provided in the form of one inch, eight channel punched tape complying with EIA-RS-227 or other input form as appropriate for the input media specified (see 6.2.1).

3.3.14.31 On-line system diagnostics. The MCU on style B machines shall perform a constant monitoring function of various activities and conditions of the control and machine. The diagnostic system shall communicate appropriate fault messages to the operator by means of a universal display. Unless otherwise specified (see 6.2.1), the on-line diagnostic capability shall determine the following fault conditions and provide appropriate error messages on the display:

- Axis feedback loop fail
- Illegal program entry
- Axis travel limits exceeded
- Axis servo motor failure
- Reader read error (on style B machines)
- Control console overheat
- Axis servo drive on/off
- Program error
- Feed limit

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3.3.14.32 Peripheral equipment interface. When specified (see 6.2.1), the control shall be equipped with an interface complying with RS-232.

3.3.14.33 Axis jog. The control panel or pendant shall have manual controls for both single and continuous, low and high speed jog movements of the slides for each controlled axis in both the positive and negative direction.

3.3.14.34 Auxiliary-functions. Unless otherwise specified (see 6.2.1), the MCU shall respond to the following program commanded auxiliary functions: interpolation, automatic cycles, and dwell. Unless otherwise specified (see 6.2.1), miscellaneous functions shall include program stop, optional program stop, end of program, spindle off, coolant on-off, end of program (rewind), and program re-start. System preparatory and miscellaneous functions shall be coded in accordance with EIA RS-274.

3.3.14.35 Tape-punch unit. When specified (see 6.2.1), style B machines shall have a tape punch unit for producing punched tapes from edited programs of tape being input and from programs stored in data memory. The unit shall have a plug receptacle and circuitry that is compatible with the tape punch output of the system, punching both paper and-mylar tapes of dimensions, characteristics and code in accordance with 3.3.14.3. The punching speed of the unit shall be not less than 75 characters per second.

3.3.14.36 Postprocessor. When specified (see 6.2.1), style B machines shall be furnished with a postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall convert specified tape information to the form and format required to operate the machine tool and control (see 6.2.1). The postprocessor shall be written for use on the type computer specified (see 6.2.1).

3.3.15 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.3.16 Drive-mechanisms. Means shall be provided to position each moving member in all axes of motion. Manual positioning shall be accomplished by utilizing handwheels and precision leadscrews or ballscrews; movement shall be initiated and controlled by the operator. Power and numerically controlled movement shall be accomplished by solid-state servo drives, electric motors, and precision leadscrews or precision, preloaded, antifriction recirculating ballscrew and nut assemblies. Numerically controlled machines shall also be equipped with position and velocity feedback devices forming a closed loop servo system. Leadscrews or ballscrews shall be hardened and precision ground to provide alignment and positioning accuracy, and be of a size, class, and precision suitable for the application.

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Leadscrews or ballscrews and nuts shall not exert excessive loads or cause premature deterioration of the feed drive motors or components. Feed rates and positions are to be monitored on numerically controlled machines by a feedback system and transmitted to the control. All slides and screws shall be shielded from contaminants by covers or guards.

3.3.16.1 Manual data input (MDI) machines. On type I, style A machines, the vertical movement of the spindle housing assembly on the column (W axis) shall be either manually or power operated, controlled by the operator. Longitudinal table movement (X axis) and cross axis saddle movement (y axis) shall be numerically controlled through MDI. On type II, style A machines, the vertical movement of the cross rail and spindle housing assembly on the columns (W axis) shall be power operated, controlled by the operator. The movement of the cross rail on the bridge (Y axis) and the longitudinal table travel (X axis) shall be numerically controlled through MDI. Axial reciprocation (Z axis), planetary grinding wheel movement (C axis), radial grinding wheel offsetting (U axis), and rotary table movement (A/D axes) shall be manually operated on both type I and type II, style A machines.

3.3.16.2 Computer numerically -controlled (CNC) -machines. On type I, style B machines, the vertical movement of the spindle assembly on the column (W axis) shall be either manually or power operated, controlled by the operator. Longitudinal table travel (X axis), cross axis saddle movement (Y axis), and planetary grinding wheel movement (C axis) shall be numerically controlled. On type II, style B machines, the vertical movement of the cross rail and spindle housing assembly on the columns (W axis) shall be power operated, controlled by the operator. Type 11, style B machines shall have numerically controlled longitudinal movement of the table (X axis), cross rail movement on the bridge (Y axis) and planetary grinding wheel movement (C axis). On both type I and type II, style B machines, rotary table movement (A/D axes), axial reciprocation of the grinding wheel (Z axis) and radial grinding wheel movement (U axis) shall be either manually or numerically controlled.

3.3.17 Position-feedback-device. The position feedback device, including all related components, hardware, and electrical equipment for the specified axes, shall be provided. The feedback device shall be of the independent type. The feedback device for each axis shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The device shall provide axis travel feedback over the entire range of axis travel of each specified axis. The position feedback device, display console and machine member shall form a closed loop system. The feedback shall be either a linear scale, tape, wire, inductosyn or rack and pinion system.

3.3.18 Standard-equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with the machine:

a. One set of special wrenches, operating and repair tools that are normally furnished by the manufacturer.

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b. Protective shield with swivel arm.

c. One worklight, adjustable to illuminate work area, complete with protective shield and on-off switch.

d. Workholding and mounting devices as follows:

(1) One set of rests for height build-up from not more than 3/8 inch to at least 9 inches.

(2) One set of "U" type clamps having adjusting screw at heel end; adjustment shall be variable through at least one inch.

(3) One set of extension bolts, complete with washers and nuts, for make-up to any height within the range of the machine.

e. The required number of indicator sets, complete with clamps, to indicate holes from 3/16-inch to 7-1/2-inch diameters.

f. Grinding wheels: shank size, number of each wheel size, type, dimensional characteristics, and composition shall be as specified (see 6.2.1).

3.3.19 Optional-equipment. The following equipment and accessories are commercially available and are considered optional. When required, the quantity of each item and its characteristics shall be as specified (see 6.2.1):

a. Storage cabinet. ,

b. Microscope.

c. Collet type adapters for grinding heads.

d. Collet sets for grinding arbors.

e. High power 9,000 RPM grinding head with removable arbors of different lengths.

f. Dust exhaust system.

g. Standard 40,000 RPM grinding head.

h. Medium speed 60,000 RPM grinding head.

i. High speed pneumatic 100,000 RPM grinding head.

j. Ultra-high speed pneumatic 160,000 RPM grinding head.

k. Precision rotary table.

l. Precision spin table.

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m. Wheel dresser, as specified (see 6.2.1).

n. Parallel set up blocks.

o. Horizontal spindle slot grinder.

p. Extension plates

q. Angle, wedge, or aligning plates.

r. Other optional equipment as specified (see 6.2.).

3.4 Sizes and-capacities. Unless otherwise specified (see 6.2.1), the sizes and capacities of the machine shall meet the requirements of table I as applicable for the machine size specified in the contract or order.

3.5 Performance. Each machine shall be capable of performing grinding operations to tolerances within +0.0001 inch on sizes 0.00005 inch TIR on roundness, and locational accuracy between holes of +0.00032 inch.

3.6 Alignment tolerance. Each machine shall meet the alignment tests in 4.6.3.

3.7 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.8 Marking of instruction plates, indicating plates and nameplates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.8.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to the machine. If a chart is furnished, it shall be placed in a transparent plastic folder or permanently sealed between clear plastic sheets with suitable means for mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.8.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer’s permanent records. The captions listed may be abbreviated, provided the entry for each caption is clear as to its identity.

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TABLE 1. Sizes and capacities,

Characteristics	Type I		Type II Styles A and B				
	Size 1	Size 2	Size 1	Size 2	Size 3	Size 4	Size 5
CAPACITY:							
Table:							
Longitudinal (X) travel (inches)	16	18	25	32	36	48	48
Cross (Y) travel (inches)	10	11	18	24	24	24	24
Work surface (inches)	23 x 15	24 x 11	30 x 24	36 x 22	40 x 33	48 x 24	52 x 33-7/8
Table top to wheel collet, adjustable (inches)	0 to 19-3/4	2 to 18	0 to 23-1/2	6 to 24-1/2	0 to 29-1/2	6 to 24-1/2	0 to 29-1/2
Permissible table load (lbs)	660	300	1100	1200	1764	2000	1764
Bridge:							
Distance between columns (uprights)	—	—	29-1/2	27	37-3/8	32	37-3/8
Spindle:							
Spindle to column guideways (inches)	13-3/8	10	—	—	—	—	—
Vertical spindle travel (inches)	4-3/4	3-5/8	5-1/2	3-5/8	7-3/32	3-5/8	7-3/32
Vertical head assembly on column (W axis) (inches)	18	12-5/8	19-1/2	13	25	13	25
Grinding capacity (inches)							
-without extension plates	<			9			>
-with extension plates	14	13-1/2	14	13-1/2	14	13-1/2	14
Radial offset (U) of grinding wheel:							
Rough adjustment (inches)	<			± .75			>
Fine adjustment (inches)	<			± .25			>
Angular adjustment from spindle centerline (for taper grinding) each side of centerline	0° to 8°	0° to 1-1/2°	0° to 8°	0° to 1/2°	0° to 8°	0° to 1-1/2	0° to 8°
FEEDS AND SPEEDS:							
Number of main spindle speeds	<			Infinite			>
Main spindle speed range (minimum to maximum) (RPM)	5 to 250	0 to 300	5 to 250	0 to 300	5 to 250	0 to 300	5 to 250
Grinding spindle speed range using full range of heads (RPM)	<			6,000 to 160,000			>
Vertical reciprocating:							
Frequency of stroke (number)	<			Infinite			>
Range (IPM)	<			4 to 240			>
Range of stroke length (minimum to maximum) (inches)	1/10 to 4-3/4	1/10 to 3-5/8	1/10 to 5-1/2	1/10 to 5-1/8	1/10 to 7-3/32	1/10 to 3-5/8	1/10 to 7-3/32
Axis (X and Y) feed rate (maximum) (IPM) (infinitely varible)	59	36	118	5	118	5	118
Machine net weight (lbs) (without electrical cabinet)	6160	3980	8800	10,000	13,200	12,000	15,400

NOTE : Machine sizes and capacities shall be not less than the stated requirements of the size ordered shown in above table. When the largest size and capacity machine shown in above table is ordered) the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE II. Alignment and accuracy requirements.

Characteristics	Maximum allowable deviation
Squareness between X and Y axes	0.0001″ per 12″
Trueness of table travel, rise and fall within	0.0001″ per 12″; 0.0002″ total
Trueness of saddle or cross carriage travel, as applicable, rise and fall within	0.0001″ per 12″; 0.0002″ total
Table flatness, checked with indicator in the spindle, measured at 5“ by 5“ grid points over entire table	0.0002″ TIR
Table sweep, at 4“ radius from spindle axis	0.0003″ TIR
Head and spindle movement square with table; 12″ high test cylinder on table, check both X and Y planes	Spindle 0.0002″ per 12″ Head 0.0002″ per 12″
Grinding test - 2" to 3" diameter hole in 6“ square meehanite block 1-1/2″ thick	Round within 0.000050″ TIR on diameter
Accuracy of measuring system	0.0001″ per 12″ 0.0002″ overall
Repeatability (at one position)	10 cycles, +0.00004"
Positioning accuracy	0.0001″ per 12″ 0.0002″ overall
Rotary table (option): 360,000 positions	
Accuracy	+2 seconds
Trueness of rotation	0.00001" TIR
Flatness of table top	0.000050″
Parallelism of table top to base	0.000075″

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.9 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).

b. Acceptance test (see 4.4).

4.3 Quality conformance inspection Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.5 and the inspection in 4.7. Failure of the item to pass the examination or inspection shall be cause for rejection.

4.4 Acceptance test. An acceptance test shall be performed on the grinder to insure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall be performed only after the grinder is installed at its final location and shall consist of the tests in 4.6. Failure of the grinder to pass one or more tests or any part of a test will be cause for rejection.

4.5 Examination. The grinder and equipment shall be examined for design, construction, material, components, electrical equipment and workmanship to determine compliance with all requirements of this specification as given in 3.1 through 3.9.

4.6 Tests. The machine shall be subjected to the following tests. All instruments, tapes, materials, and tools required to perform and evaluate these tests shall be furnished by the supplier.

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4.6.1 Operational test. The machine shall be operated at no load for not less than one hour. The machine and the numerical control system shall be operated in accordance with the manufacturer's standard operating test procedure for warm-up and run-off checks. During the warm-up period, proper operation of all controls, motors, adjustment mechanisms, and accessories shall be verified. After initial warm-up for a period of not less than four hours, the machine shall be cycled continuously under numerical control to verify its ability to receive and perform all program functions without error. This operation shall include spindle speed and feed rate changes, rapid traverse of all slides, simultaneous movement of slides, and automatic feed cycles, as applicable. The numerical control shall also be tested for proper operation of MDI, program edit, and system diagnostics. Should a malfunction occur, it shall be corrected and the operation test repeated until a full four hours of running time is completed without failure.

4.6.2 Performance -tests. The following test shall be performed to demonstrate the machine's capability to accurately position, grind and relocate. Prebore three 1.005 inch diameter through holes in an equilateral triangular pattern in a metal plate not less than one-inch thick. The plate shall be of a material chosen by the supplier. One or both faces of the plate may be surface ground prior to the test. The dimensions between bored holes shall be within one inch of the limits of cross-axis travel of the sliding members. Finish grind the holes to a diameter of 1.0170 inch. Maximum deviation on size shall be not more than 0.0001 inch. Remove the workpiece from the machine and relocate it on the worktable, rotated 90 degrees. Using a locating dial indicator and the measuring system of the machine, relocate the ground holes. The machine shall accurately relocate the second and third holes to within plus or minus 0.00032 inch. The machine's positioning shall be determined by comparison to a linear measuring instrument with calibrated accuracy directly traceable to the National Bureau of Standards.

4.6.3 Alignment and -accuracy tests. Each machine and its components shall be tested in the assembled machine for alignment and accuracies. Unless otherwise specified (see 6.2.1), the alignment and accuracies shall meet the requirements in table II. The instruments and and methods used in performing the test shall be at the option of the supplier provided they accurately determine the tolerances of the machine. Tests shall be performed in an ambient temperature which differs from 70° F by not more than 5°.

4.6.4 Overshoot test. Unless otherwise specified (see 6.2.1), overshoot shall not exceed ().0004 inch at any feed rate up to 20 percent of the maximum.

4.6.5 Electromagnetic -interference -control -tests. Equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.2.10.

4.7 Packaging -inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

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5. PACKAGING

5.1 Preservation, packaging, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall conform to the requirements of ASTM-D-3951. When specified (see 6.2.1), level A packaging, level A or level B packing, and special marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended-use The jig grinding machines covered by this specification are intended for use in any manufacturing shop where precision grinding of jigs, dies, or other similar precise work is performed. They will be used to locate and grind cylindrical holes, conical holes with tapers in either direction, and contours including combinations of radii, tangents, angles, and flats.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, style, and size required (see 1.2).

c* Units of measurement system to be used in the design and construction of the machine (US, SI, or other system of measurement) (see 3.1.1).

d. If measuring and indicating devices are required to be graduated in a specific measurement system, state required system (US, SI, or other system of measurement) (see 3.1.1.1).

e. Additional safety and health requirements, if required (see 3.1.5 and 6.3).

f. Painting, if different (see 3.2.5).

g. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.2.10).

h. Grinding wheel shank sizes 1/4 and l/8-inch, if required (see 3.3.3).

i. Numerically controlled grinding wheel off-setting (U-axis), if required (see 3.3.3.2).

j. Numerically controlled spindle reciprocation (Z axis), if required (see 3.3.3.3).

k. Number, size, and configuration of T-slots, if different (see 3.3.4).

l. Precision rotary table or precision, tilting rotary table, if required (see 3.3.4.1).

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m. Size of rotary table, if required (see 3.3.4.1).

n. Tailstock for tilting rotary table, if required (see 3.3.4.1).

o. Grinding wheel requirements, if different (see 3.3.8).

p. Coolant system, if not required (see 3.3.10).

q. Pneumatic system, if not required (see 3.3.12).

r. Electrical standards, if different (see 3.3.13).

s. Motor enclosure, if different (see 3.3.13.1).

t. Axes under numerical control (see 3.3.14.2).

u. Parabolic interpolation, if required (see 3.3.14.2).

v. Input media for the control, if different (see 3.3.14.3),

w. EIA-RS-244 in lieu of EIA-RS-358 (see 3.3.14.3).

x* Tape reader reels, if different (see 3.3.14.3).

y. Data display, if different (see 3.3.14.4).

z. Memory type, if different (see 3.3.14.9).

aa. Part program storage capacity, if different (see 3.3.14.9).

bb. Standby battery power required, if different (see 3.3.14.10).

cc. Mirroring in four quadrants, if required (see 3.3.14.14).

dd. Additional NC features, if required or if different (see 3.3.14.25).

ee. Operator control panel functions, if different (see 3.3.14.26).

ff. Pendant control panel, if required; pendant control functions, if different (see 3.3.14.26.1).

gg. MCU programs input and storage media, if different (see 3.3.14.30).

hh. On-line diagnostic fault conditions, if different (see 3.3.14.31).

ii. Peripheral equipment interface, if required (see 3.3.14.32).

jj. Miscellaneous and auxiliary functions, if different (see 3.3.14.34).

kk. Tape punch unit, if required (see 3.3.14.35).

11. Postprocessor, if required; specify computer name, model, form and format (see 3.3.14.36).

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mm. Hour meter, if required (see 3.3.15).

nn. Standard equipment, if different (see 3.3.18).

oo. Size, number, type, characteristics and composition of grinding wheels, if required (see 3.3.18.f).

PP. Optional equipment, if required or if different (see 3.3.19).

qq. Type of wheel dresser, if required (see 3.3.19.m).

rr. Size and capacities, if different (see 3.4).

ss. Repair parts, if required (see 3.7).

tt. Lubrication chart, if different (see 3.8.1).

uu. Nameplate, if different (see 3.8.2).

vv. Quality conformance inspection, if different (see 4.3).

ww. Acceptance test, if different (see 4.4).

xx. Alignment and accuracy test requirements, if different (see 4.6.3).

yy. Overshoot, if different (see 4.6.4).

zz. Preservation, packaging, and marking, if different (see 5.1).

6.2.2 Contract -data-requirements. All data requirements (such as parts lists, wiring diagrams, ' operating manuals, maintenance manuals, and alignment and test results) shall be required on a DD Form 1423, Contract Data Requirements List, and shall be furnished in accordance with each specified DD Form 1664, Data Item Description, as shown on the DD Form 1423.

6.3 Safety-and -health-requirements . Paragraph 3.1.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat . Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment), the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all points-of-operation guarding as required by the manufacturer. The procuring activity should specify the exact configuration of the guard required if, as in most cases, the guard configuration is dependent on the size and configuration of the workplaces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.l(e).

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Custodians:
Air Force - 99
Army - AL
Navy - SH

Review activities:
Air Force - 84
Army - AV
DLA - GS
Navy - SH

User activities:
Air Force - 99
Army - None
Navy - MC

Preparing activity:
DLA - IP

Project 3415-0154

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80267	Grinding Machines, Jig, Vertical, CNC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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